LLM-Integrated Satellite Image Analysis and Change Detection System

MAJOR PROJECT-IIREPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

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KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

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CERTIFICATE

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	LLM-Integrated Satellite Image Analysis and Change Detection System
2	**Student Names & Numbers**	**Greeshma Kandregula(228W1A12F9)** **Teja Gavara(228W1A12E9)**
3	**Name of The Guide**	**Dr.V.Radhesyam**
4	**Research Group**	Deep learning, remote sensing, and web technologies.
5	**Application Area**	Urban Planning and Development, Enhanced Environmental Monitoring.
6	**Aim of the Project**	To analyze satellite images for classification, segmentation, and change detection.
7	**Project Outcomes**	• Accurate classification of land-cover types using EfficientNet • Precise segmentation of satellite images using SAM • Reliable change detection using U-Net • Real-time satellite data retrieval using NASA GIBS • Interactive visualization of results through a web dashboard • LLM-based explanations and automated report generation

Student Signatures

1.

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Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr.M.Suneetha**, Dean R&D, Professor &Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr.V.Radhesyam, Associate Professor** for **his** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficulty on my part to finish the project work.

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ABSTRACT

This intelligent system is designed to make use of satellite image analysis with deep learning, real-time data collection, and explainable artificial intelligence. These features will enable land cover classification, segmentation, and change detection in one system. Satellite images will be collected either through upload or by means of dynamic retrieval from NASA GIBS based on geographic and time inputs. For segmenting the images accurately, SAM will be deployed in the system. For classification, efficientNet will be used to provide accurate and computationally efficient classification of land covers. For change detection, the U-Net model will be applied to create maps showing changes at various time intervals. Moreover, there will be an interactive dashboard that displays before and after images along with segmentation and change detection results. In addition to the above, an LLM will be added to generate natural language explanations and answers and create reports in a structured manner. In conclusion, this system will provide an automated and interactive tools for geospatial analysis.

Keywords: Satellite Image Analysis, Land-Cover Classification, Change Detection, EfficientNet, Segment Anything Model (SAM), U-Net, Remote Sensing, Geospatial Analysis, NASA GIBS, Deep Learning, Image Segmentation, LLM-based Interpretation

CHAPTER-1

Introduction

1.1 Origin of the Problem

As a result of the growing trends in urbanization, industrialization, and pollution, a remarkable alteration in LULC has taken place. Such alterations should be monitored in order to attain sustainable development, efficient responses to disasters, and proper environmental management. However, the traditional techniques for analyzing satellite images are characterized by high costs and the lack of speed as well as scalability to vast geographical areas [1].

With the improved technology of remote sensing systems, there is now an abundance of satellite images, ranging from Sentinel to Landsat. Even though such information is highly detailed and comprehensive, processing satellite data into useful results poses a great difficulty. According to a study, the inconsistencies in satellite data regarding spatial resolution, atmospheric effects, and seasonal changes can affect the precision of the land cover classification [2].

In order to overcome these problems, the deep learning approach has been extensively applied to automated image interpretation. Studies show that CNNs and other modern deep learning algorithms enhance the classification process by incorporating complicated spatial attributes [3], [4]. Additionally, AI methods have facilitated the automated detection of changes in land cover without relying on human interpretation [5].

Latest studies have considered multi-temporal analysis and deep learning models to detect environmental changes like urbanization and vegetation degradation [6], [7]. Machine learning algorithms and hybrid models have been investigated for use in complicated terrain areas; however, their efficacy may differ based on the variety of landscapes [8].

Further improvement in classification accuracy and feature extraction has been achieved through more sophisticated architectures such as the transformer model and semantic segmentation algorithms [9], [10]. Fine-grained classification has become

more accurate due to the use of high-resolution images; however, the processing requirements have also risen [11]. Hybrid solutions integrating deep learning and optimization algorithms have also been suggested to increase efficiency and accuracy [12]. Some emerging technologies, such as quantum-classical neural networks, provide insight into the future development direction [13].

However, some issues have yet to be addressed, including computational costs, non-intuitive algorithms, and weak integration of real-time analysis and user interface design. Consequently, the need for a smart and efficient solution that integrates deep learning and change detection for insightful analysis of satellite imagery is evident.

Practical difficulties include, but are not limited to, cloud cover, seasonality, and the problem of non-uniform spatial resolution, which affects the effectiveness and efficiency of data classification. In addition to that, the current available solutions have limitations in terms of real-time operation and the ability to provide a user-friendly interface. It is therefore imperative to devise an intelligent solution that will address these concerns.

1.2 Basic definitions and Background

1.2.1 Basic definitions

The interpretation of satellite images and land cover mapping involves an understanding of core concepts related to remote sensing, deep learning, and geospatial analysis.

Land Use and Land Cover (LULC)

Land Use and Land Cover (LULC) are two terms that are the basic concepts behind remote sensing and environmental science. Land use refers to the use of land for certain activity carried out by man, which can include agricultural land use, urbanization of land, industrial usage, transportations, and other kinds of activities. This concept focuses more on how man utilizes his land resources in order to meet his needs. On the other hand, the concept of land cover relates to the nature of surface of the land, which includes forests, water bodies, soil, vegetation and construction areas.

Both these concepts need to be distinguished from each other because the same land cover does not necessarily mean that the land uses have to be the same. LULC plays an important role in environmental management and deforestation detection.

Remote Sensing : According to the definition of remote sensing, it is the science or art of observing objects from a distance through the use of sensors mounted on platforms such as aircraft or satellites and collecting information in the form of energy that is emitted or reflected back by the earth.

There are many advantages associated with remote sensing. The first advantage of remote sensing is the efficient coverage of large areas that are hard to access. Remote sensing involves collecting information at different wavelengths and times, thus giving useful information on changes in land cover, vegetation, water, and atmosphere. Remote sensing is used in many disciplines such as environmental management, agriculture, weather forecasting, and military operations.

Satellite Images: Satellite images are images of the earth's surface obtained by utilizing various types of sensors mounted to the satellites. The sensors include those used in Sentinel satellites and in the Landsat missions. Satellite images provide an extensive range of spatial and temporal data, hence allowing for constant monitoring of the earth's surface.

A satellite image consists of a variety of spectral bands which include visible as well as invisible radiations such as red, green, blue, infrared, and thermal radiations. Through the use of spectral bands, materials can be distinguished by their reflection and absorption rates of light radiations. For example, plants reflect a great deal of near-infrared radiations while water absorbs almost all radiations.

Image Segmentation: Image Segmentation can be defined as an integral part of the Image Processing process, wherein the image is segmented into multiple segments, which can be considered homogenous and are semantically relevant. As far as satellite images are concerned, the process of segmentation enables the division of regions, which helps in the recognition of various objects within the imagery, including vegetation, water bodies, roads, etc.

In general, Segmentation refers to the process of simplifying the image by categorizing it on the basis of pixel features, such as color, intensity, texture, etc. The latest segmentation methodologies make use of Convolutional Neural Networks, including U-Net and Mask R-CNN for Satellite Image Segmentation.

Segmentation plays a very vital role when it comes to land cover classification and object detection tasks.

Image Classification: Image Classification is one of the most significant applications of computer vision and involves assigning an appropriate label to an image or segments of the image depending on the content contained in the image. The application of image classification in satellite imagery is concerned with identifying various types of land covers like vegetation, bodies of water, urban areas, and bare soil.

The image classification techniques used in the past involved manually designing image features. However, modern machine learning technology has allowed automatic feature design. CNN has been proven to be quite effective when dealing with image classification problems due to its proficiency in extracting features from massive amounts of data. EfficientNet and ResNet are some of the most commonly used neural networks in image classification. These networks are quite efficient in operation owing to their hierarchical nature.

Image classification helps in the monitoring of the environment, agricultural activities, and urbanization by means of land cover assessment.

Change Detection: The term Change Detection implies the identification and interpretation of changes taking place in the condition of an object or phenomenon through analysis of images obtained by the satellite in different periods of time. This is one of the primary applications of remote sensing. Change Detection plays a crucial role in determining the changes taking place in the specific area due to natural as well as human activities.

Change Detection can be utilized in detecting the change in land cover in specific areas including deforestation, urbanization, desertification, or any kind of change taking

place in that area. Technologies used for Change Detection include advanced technologies like Deep Learning and Image Differencing techniques.

NDVI (Normalized Difference Vegetation Index): This is an index used to determine the condition of the vegetation through spectral analysis. The normalized difference vegetation index utilizes the reflectance differences of both near infrared and red bands from the satellite data imagery. When the reflection of the near-infrared band increases, it indicates that there is dense vegetation in the area.

The NDVI has a range between -1 and +1, whereby a positive result indicates dense vegetation, while a moderate one indicates sparsely vegetated areas, and negative results indicate barren lands, cities, or water bodies.

The NDVI index can be used in several industries such as agriculture and forestry.

Deep Learning: Deep learning is a sophisticated way to implement the machine learning approach by using artificial neural networks that comprise many layers to detect complex feature patterns in the data set input. Such techniques are capable of learning hierarchical patterns autonomously; therefore, they can be applied to process large data sets with numerous features, such as satellite imagery.

Currently, various deep learning techniques are implemented in the field of geospatial technology to address certain issues associated with image classification, segmentation, and change detection. Contemporary deep learning architectures include CNN, U-Net, and Transformers; these models provide better results compared to older techniques.

This potential of deep learning has led to a revolution in the analysis of satellite images.

Large Language Models (LLMs): LLMs are AI models that are able to process, analyze, and produce text content similar to how humans communicate.They have been developed by training on large data sets of texts and can carry out several functions related to natural language processing. The functions include but are not limited to generating text summaries, translations, and answering queries among others.

GPT models can interpret the outputs and provide useful information to the end-user. In the suggested system, the use of LLMs is aimed at enhancing user friendliness via

interpretation of the outputs produced by the model, generation of text summaries of the outputs, and understanding the satellite image analysis.

1.2.2 Background of the Study

With improvements in resolution of satellite imagery and developments in computation technology, there have been considerable changes in the geospatial analysis sector in recent years. The old methods used in geospatial analysis involved manual processing and application of traditional machine learning algorithms. However, with the emergence of deep learning models like Convolutional Neural Network (CNNs), significant progress has been achieved in the field of land cover classification.

The current techniques involve efficient algorithms such as EfficientNet, ResNet, and segmentation networks that help extract valuable information from satellite images. Furthermore, the use of multi-temporal information obtained through the use of satellites has led to an effective change detection procedure, allowing the detection of all kinds of environmental changes.

Another set of recent developments includes the implementation of various approaches, including segmentation, classification, and clustering techniques. The implementation of various environmental indices, such as NDVI and NDWI, is also one of the new trends in the field. Nevertheless, there are some barriers to the successful implementation of such technologies, including heterogeneity of input data, high computational power required, and low interpretability. To overcome these barriers, modern AI systems have introduced logic reasoning.

Using large language models, it becomes possible to make human-like interpretations and predictions of events. Thus, the background of the presented research involves the evolution from manual approaches to intelligent automated systems in terms of image analysis.

The development of the analysis of satellite images may be generally divided into four phases: manual analysis, machine learning techniques, deep learning systems, and smart AI-based techniques. Traditionally, the first step included extensive human involvement, which was later succeeded by techniques based on machine learning. The

next step was driven by developments in deep learning that helped automate feature extraction and improve performance. The most modern solutions incorporate deep learning and AI/LLMs to ensure explainability.

1.3 Problem Statement

1.3.1 Problem Statement

The factors like urbanization, deforestation, and environmental changes always affect the features of the land use/land cover in the world. Monitoring of the above mentioned problems is vital for achieving sustainability goals. However, the existing techniques of classifying satellite imagery are complicated, require professionals, and can hardly be used on a larger scale [1].

Given the amount of satellite data available via Sentinel and Landsat, an automated analysis is needed now more than ever before. Even though deep learning systems have shown improvements in terms of accuracy, problems such as variability of satellite imagery based on different weather conditions and other factors persist [2], [8].

Current systems are focused more on classification or change detection, without any single approach that incorporates segmentation, classification, and time component simultaneously. Besides, most of the models rely on extensive training data and computationally heavy processes, which make them difficult to utilize in practice [3], [4]. Although some modern models, like transformers and hybrid networks, perform better than older ones, they suffer from a lack of interpretability and a user-friendly interface [9], [10].

In addition, most approaches are limited by the scope of information provided; specifically, environmental indices and vegetation/water health assessment are usually not covered by existing models. The lack of visualization capabilities and natural language processing adds up to this issue.

Therefore, it is necessary to build a smart, automated, and scalable model for integrating segmentation, classification, change detection, and environmental analysis, along with other quantitative models and explanations.

1.3.2 Objectives

The main objective of research is the design of Intelligent, Automated Satellite Images Analysis System using Deep Learning, change detection and Explainable AI. The objectives of this study include:

1. **Designing and Implementation of an Integrated Satellite Image Analysis System:** To design and implement a satellite images analysis system which includes various stages in acquisition of satellite images such as Acquisition, Preprocessing, Segmentation, Classification, and Change Detection of satellite images.
2. **Facilitating Geographic Region Selection:** To design a system that facilitates selection of geographic region via the use of map.
3. **Segmentation and Classification of Land Covers:** For performing the tasks of segmentation and classification of land covers like vegetation, urban areas, water and roads, it is essential to use different types of Deep Learning Models for example SAM, Efficient Net and ResNet.
4. **Efficient Change Detection Algorithms:** In order to facilitate implementation of efficient change detection algorithms in addition to deep learning algorithms such as U-Net, some traditional algorithms K-Means, Image Differencing should be included in the system.
5. **Quantitative and Comparative Analysis**: For performing quantitative and comparative analysis such as Vegetation loss percent and Urbanization rate etc.
6. **Include Interpretation and Reporting via LLMs:** To increase the usability of the system by including Large Language Model-based reporting that will help in providing explanation and answer user questions.
7. **Make Sure That The System is Flexible and Can Accept Real-Time Satellite Data:** To make sure that the system can be flexible in accepting either uploaded images or real-time satellite data via API services like NASA GIBS.

1.3.3 Outcomes

The suggested system will be capable of providing an all-inclusive intelligent solution for analyzing satellite images by combining sophisticated deep learning algorithms,

environmental indices, and explainable AI methods. The following are the primary anticipated results of this research project:

1. **High-Accuracy Land-Cover Classification:** The solution should provide high classification performance in detecting land cover classes including vegetation, urban area, water body, and road using the EfficientNet architecture. Further enhancement in classification accuracy will be provided by the application of the EfficientNet architecture on segmented areas.
2. **High-Quality Image Segmentation with SAM:** With the implementation of the Segment Anything Model (SAM), it will be possible to carry out image segmentation with high accuracy.
3. **Change Detection with U-Net:** The system will be able to perform temporal change detection between satellite images utilizing deep learning algorithms. U-Net will be used to mark regions that have undergone changes, as well as calculating statistics like percentage of change.
4. **Real-Time Access to Satellite Images:** Real-time access to the satellite data is achieved by integrating NASA GIBS and allowing users to enter latitude and longitude along with a time frame to receive up-to-date data.
5. **Natural Language Understanding with LLM Integration:** LLM integration allows the system to automatically produce natural language reports, summaries, as well as predictive analyses.
6. **High System Efficiency at Scale**: The system will be designed to work with big data and satellite data efficiently to ensure reliable results in the real world.

1.4 Applications

The satellite image analysis system has multiple applications across various fields, owing to its capability to classify land cover accurately and intelligently interpret geospatial information.

1. **Urban Planning and Smart Cities**: This system can be used to identify urbanization trends and analyze land use in cities. With the help of satellite images, this tool can be employed to make informed planning and zoning decisions by urban planners.

2. **Disaster Impact Analysis**: The system can also be used to understand and mitigate the impact of natural disasters. The application of the system is helpful in making damage assessments and planning recovery plans based on pre and post-disaster satellite imagery.
3. **Climate Change Assessment**: Through the use of the tool, long-term environmental trends can be assessed and researched upon to formulate policies related to climate change. This includes assessing impacts such as urbanization, deforestation, and altered waterways.
4. **Geospatial Analysis and Decision Support System**: Researchers and policymakers can utilize the tool to conduct large scale analysis of geospatial data and gain valuable insights. The inclusion of LLM-based explanations in the system improves analysis and decision support.

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

The literature on satellite image analysis demonstrates significant progress in land-cover classification, deep-learning architectures, and change-detection frameworks. The following section reviews six key systems that form the foundation for this project.

1. Deep Learning for Urban Land Use Classification – Li et al. (2024) [1]

A deep learning approach was suggested by Li et al. for high-resolution satellite imagery used in urban land use analysis. This research explores several CNN models for spatial feature extraction to enhance classification accuracy in urban areas. It also demonstrates improved performance compared to traditional methods by effectively capturing complex urban patterns.

Advantages: Achieves high classification accuracy, efficient feature extraction technique, applicable for big data sets.

Disadvantage: Large training data set and high computational power required.

Relevance: Provides justification for using EfficientNet for land cover classification in this project.

2. Sentinel-2 Land Cover Classification Review – Jocea et al. (2025) [2]

Jocea et al. provided a systematic review of land cover classification approaches based on Sentinel-2 imagery, covering traditional methods as well as those based on deep learning. Focus is made on real-life obstacles when implementing such techniques, such as complexities associated with preprocessing.

Advantages: Systematic approach, practical lessons learned, multi-spectral image processing.

Disadvantage: No clear guide on how to implement such an approach in practice.

Relevance: Helps realize what problems can arise when working with satellite images.

3. Deep Learning for Land Use Classification using HS-LiDAR – Rehman et al. (2025) [3]

Rehman et al. proposed a deep learning method that utilizes the hyperspectral and LiDAR information for land use classification. This combination of spatial and elevation information contributes to increasing the efficiency of the classification process in complicated terrains.

Advantages: Accurate results, multiple information sources, increased feature quality.

Disadvantage: Data specific requirement and computationally intensive.

Significance: Emphasizes the role of effective feature extraction in classification.

4. **Deep Learning-Based Satellite Image Segmentation – Deressu et al. (2025) [4]**

Deressu et al. presented a method of segmenting images and classifying them into different categories of land covers using deep learning. This technique enhances pixel-level accuracy through segmentation and classification.

Advantages: Highly accurate segmentation results and enhanced pixel-level analysis.

Disadvantage: It is computationally expensive and ineffective for detecting smaller objects.

Relation to current project: Validates the application of SAM in this project.

5. AI and Remote Sensing for Land Cover Change Detection – Gu & Zeng (2023) [5]

Gu and Zeng reviewed the application of artificial intelligence in the detection of land cover changes from satellite data. The study highlights the transition from traditional methods to AI-driven automated systems for detecting environmental changes.

Strengths: Automation, improved efficiency, wide applicability.

Limitation: Limited interpretability and lack of explainable outputs.

Relevance: Justifies the integration of AI-based change detection in this project.

6. Multi-Temporal CNN-Based Change Detection – Şimşek (2025) [6]

Şimşek introduced a CNN-based technique employing PlanetScope satellite imagery from multiple times to identify land cover changes.

Advantages: Effective time series analysis, improved accuracy in change detection.

Disadvantage: Effectiveness of the algorithm relies heavily on the quality of the data and pre-processing processes.

Relevance: Provides justification for the use of deep learning-based change detection (U-Net) in the project.

7. Transformer-Based Land Use Classification – Khan et al. (2024) [9]

A transformer deep learning approach was introduced by Khan et al. to classify land use and land cover types using satellite imagery. This technique makes use of self-attention methods to extract features based on the spatial relationship within the data, thus producing better feature representation compared to the ones created through CNN models.

Advantages: It captures global spatial relationships and achieves high classification accuracy through improved feature extraction. Attention mechanisms also enhance explainability by revealing how decisions are made.

Disadvantage: It requires significant computational resources and has a more complex architecture compared to CNN-based models, making it harder to implement and deploy efficiently.

Relevance: This study illustrates the capabilities of advanced architectures beyond CNNs Also, the importance of interpretability is highlighted, which agrees with the interpretability approach used in the present research.

2.2 Summary of Literature Study

From the point of view of the literature review, there has been remarkable progress with regards to satellite image interpretation. In particular, LULC classification and change detection have been substantially advanced using the potential of deep learning technology. Initially, conventional machine learning was used, but its capability was somewhat restricted in capturing intricate spatial relationships. Nevertheless, a number of studies show that CNNs and other deep learning techniques enable highly accurate satellite image classification owing to the extraction of higher-order spatial attributes [1], [3].

Moreover, several papers emphasize the significance of multi-spectral and multi-temporal satellite data sources like Sentinel-2 and PlanetScope, which provide better results in terms of LULC classification and change detection [2], [6], [7]. The use of segmentation methods has also proved to be effective in increasing pixel-wise accuracy

and achieving reliable land cover discrimination [4], [10]. Furthermore, ensemble models and hybrid deep learning methods have been studied [7], [12].

The latest research focuses on the use of modern architectures like transformer-based models and quantum-classical neural networks that facilitate enhanced feature extraction and classification abilities [9], [13]. Fine classification techniques through the use of high-resolution images have made accurate classification possible; however, at the cost of increasing computational complexity [11].

Additionally, the use of environmental indices and spectral features has proven effective in improving classification efficiency through contextual knowledge acquisition [15]. AI solutions for detecting changes within the environment have made the process easier while also being efficient [5].

However, there still remain certain challenges in this domain. Models require massive amounts of labeled data and computational resources, hence resulting deployment hard in practical applications [3], [4]. Changes in environmental indices and geographical diversity affect the performance of these models as well [2], [8]. Moreover, the majority of these systems do not possess interpretability or user-friendly interfaces [10].

This means that, from the existing literature, there is a need for a system that will be integrated to ensure that classification, change detection, environmental impact assessment, and explainability have been achieved. This will form the basis of the new system that will overcome the above-mentioned limitations through deep learning and LLMs.

Additionally, such a system should emphasize scalability and real-time processing capabilities to ensure its applicability across diverse geographic regions and dynamic environmental conditions. Furthermore, incorporating user-friendly interfaces and visualization tools would enhance accessibility for non-expert stakeholders and support informed decision-making.

Table 2.1: Summary of Prior Work

Author (Year)	Model	Dataset	Accuracy	Key Insight
Li et al. (2024) [1]	CNN-based DL Models	Satellite imagery (Urban datasets)	High (~85–90%)	CNNs can efficiently detect spatial characteristics for land-use classification in urban areas
Jocea et al. (2025) [2]	Multiple ML/DL Methods	Sentinel-2	Varies (~80–90%)	Precise classification is influenced by the quality of pre-processing and implementation in reality.
Rehman et al. (2025) [3]	Deep Learning (HS + LiDAR)	HS-LiDAR datasets	High (~88–92%)	Data obtained from multiple sources significantly enhances the classification process
Deressu et al. (2025) [4]	Segmentation DL Models	Satellite imagery	~85–90%	Segmentation enhances pixel-level classification accuracy.
Gu & Zeng (2023) [5]	AI-based Methods	Remote sensing datasets	Moderate–High	Automatic identification of land cover changes is possible with AI.
Şimşek (2025) [6]	CNN Model	PlanetScope images	~87–91%	Multi-temporal data helps in better change detection.
Wang et al. (2025) [15]	DL + Spectral Indices	Satellite Imagery	~90–93%	Spectral indices (NDVI) enhance classification performance.

2.3 Software Requirement Specification

The Software Requirements Specification (SRS) will outline the functionalities and non-functionalities of the satellite image analysis system. The SRS will describe the software environment and requirements needed for efficient development and deployment of the system.

2.3.1 Software Requirements

The system is developed by utilizing a range of contemporary technologies related to satellite image processing, deep learning, and user interaction on the web.

1. **Programming Language:** The primary programming language is Python because of its broad application for machine learning, image processing, and working with APIs.
2. **Libraries and Frameworks**:
    a. TensorFlow/PyTorch – TensorFlow and PyTorch are used to create deep learning architectures like EfficientNet, U-Net, and Segment Anything Model.
    b. **OpenCV –** OpenCV is applied to preprocess images and perform other image processing actions.
    c. **NumPy & Pandas –** These libraries are used to manipulate numbers and work with data.
    d. **Scikit-learn –** The Scikit-learn library is employed for preprocessing images and auxiliary machine learning purposes**.**
3. **Web Framework**: Flask is employed for creating REST APIs and interacting with backend processes.
4. **Frontend Technologies**: These technologies are used to build an interactive interface.
    a. **Leafmap** technology is used to select ROIs by map.
    b. **Satellite Data Sources**: NASA GIBS WMS service is used to download satellite images in real time.APIs are used to access Sentinel and Landsat data.
    c. **LLM Integration**: APIs from the LLM services are used to create explanations and generate reports.

2.3.2 Functional Software Requirements

1. **Input Handling:** The system must handle input in the following two ways:
   - Image upload via satellites
   - Geographic coordinates and date range as inputs
2. **Retrieval of Satellite Images**: The system must fetch satellite images based on user inputs by using APIs like NASA GIBS.
3. **Image Segmentation**: The image segmentation should be done through the use of SAM for the identification of regions in the images.
4. **Land-Cover Classification**: The classification should be done on each of the segments by use of EfficientNet.
5. **Change Detection**: Detection of change is to be done using U-Net between two different periods and highlighting any changes.
6. **Visualization of Results**: The system should present outputs such as:
   - Pre-satellite images and post-satellite images
   - Change detection maps
   - Segmentation output
   - Summary tables
7. **LLM-Based Interactions:** The system should offer a chatbot interface to provide explanations for results, answer questions, and create reports.

2.3.3 Non-Functional Requirements

Non-functional requirements define system quality attributes and constraints.

1. **Performance**: The system must be able to perform high processing of satellite images without any latency, especially if the processing needs to be done in real time.
2. **Scalability**: The system must also have good scalability when there are larger amounts of data and larger geographic regions.
3. **Accuracy**: It is very important for the system to have high accuracy in its classifications and segmentations.
4. **Ease of use**: It is also important for the system to be easy for users to use.

5. **Reliability**: It is also necessary for the system to be reliable and not have any problems.
6. **Security**: It is very important for all user data to be secured and confidential.
7. **Maintainability**: It is also important to consider maintenance while designing the system.

CHAPTER-3

Proposed Method

3.1 Design Methodology

This process of developing the system involves a very well-defined and modular approach in order to obtain efficient image analysis, land cover classification, and changes in the landscape. All aspects of data acquisition, preprocessing, segmentation, classification, environment analysis, and intelligent interpretation have been incorporated into the system for efficient processing.

1. **General System Workflow*:*** The system works through a series of interconnected modules that start with user input and conclude with visualization and intelligent interpretation of results.

   Steps in the Workflow:

   User Input → Data Acquisition → Preprocessing → Segmentation → Classification → Change Detection → Environment Analysis → Visualization → Intelligent Interpretation using LLMs → Reporting

2. **User Input Acquisition:** The system receives input from the user in two versatile ways:

   a. **Image-based input**: Before and after satellite images are uploaded by the user.
   b. **Coordinate-based input:** Geographic coordinates and timespan are specified by the user.

3. **Satellite Data Acquisition:** When the input is coordinates-based, the system uses the satellite data obtained through NASA GIBS (WMS). It:

   a. Acquires images according to geographical coordinates and selected days
   b. Utilizes different satellite layers (MODIS Aqua, Terra, VIIRS, Blue Marble)

c. Provides alternative data access options when the desired satellite data cannot be found

4. **Image Segmentation using SAM**: In the proposed system, the segment anything model (SAM) is employed to perform segmentation operations on the obtained satellite images.

   Procedure:

   a. Pass the input image through SAM
   b. SAM outputs the segmented mask corresponding to different segments
   c. Each segmented region represents different land covers

   The process of segmentation helps in increasing the accuracy of classification by isolating the required regions from the rest of the image.

5. **Land Cover Classification using EfficientNet:** Once the image segmentation operation is completed, each segmented area is further classified using the EfficientNet algorithm.

   Procedure:

   a. Feed the segmented region to EfficientNet for feature extraction
   b. The extracted features are assigned appropriate classes, such as vegetation, urban area, water, etc.
   c. Output confidence scores corresponding to each classification operation

6. **Change Detection through U-Net**: Change detection is achieved through an advanced deep learning architecture based on U-Net.

   a. Two images are combined to form a six-channel image input
   b. U-Net analyzes the input to produce a binary change map
   c. Changes are indicated in the generated map

7. **Visualization and Output Rendering:** The system provides its results via an interactive dashboard containing:

a. Pre-and post-satellite images
b. Maps showing change detection (regions marked)
c. Segmentation output
d. Land classification results with label tags and confidence score
e. Land cover percentage table

8. **Interpretation via LLM:** The system also features a component that uses Large Language Models (LLMs) to make results usable and explainable.

Main functions:

a. Produce human-like language descriptions of results
b. Respond to inquiries related to analysis
c. Generate predictions
d. Create tabulated reports

9. **Error Management and Robustness:** The system is designed to provide high reliability through features like:

a. Data validation
b. Missing satellite image handling
c. Falling back on alternate layers when necessary
d. Retrying if date is unavailable

10. **Advantages of Proposed Methodology**: The proposed methodology offers several advantages, including modular design, scalability, and integration of advanced deep learning models. Accuracy can be improved using segmentation and classification techniques, while applicability can be increased by incorporating real-time satellite data. Additionally, the integration of LLM enables explainability, making the system accessible to a wider audience.

3.1 Actors and Use Cases

Satellite Image Analysis System's use case diagram shows the relationship between actors outside the system and its functionality represented by use cases. The use case diagram in Figure. 3.1 shows how various users interact with the system in order to analyze and interpret satellite images and detect changes.

3.1.1 Actors

1. **User (General User):** The User works with the web interface to carry out satellite image analysis through the system. The User has the ability to choose the area of interest, input images or coordinates, and get analysis output. This actor includes any person desiring simple geospatial information without having special knowledge.

2. **GIS Analyst**: The GIS Analyst works with the system to conduct geospatial analysis and interpretation. He focuses on conducting land-cover classification analysis and segmentation analysis among other environmental changes. These outputs are used by the actor in practical uses such as mapping and environmental monitoring.

3. **Researcher**: This actor is interested in studying trends, evaluating output models and analyzing environmental patterns through the use of the system.

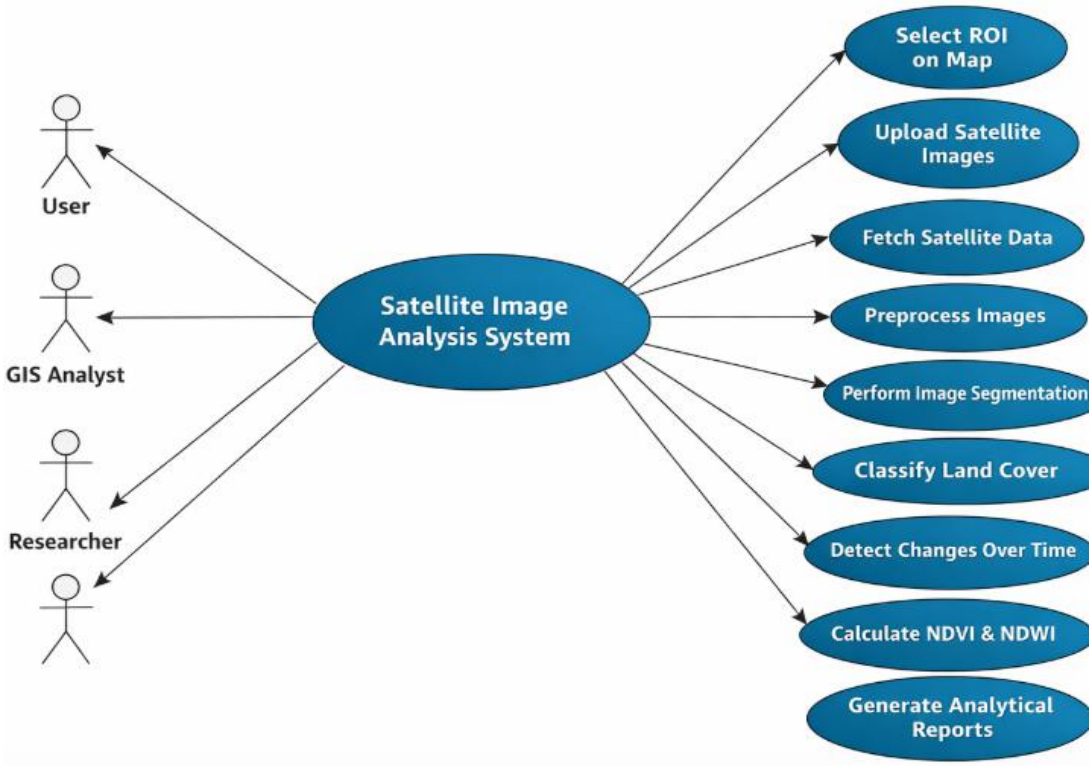

Figure-3.1: Use-case view of the Satellite Image Analysis System

3.1.2 Use Cases

1. **Select Region of Interest (ROI):** The user can select the region of interest using a map interface.
2. **Upload Satellite Images**: Users are allowed to upload pre- and post-satellite images.
3. **Obtain Satellite Imagery**: The system extracts satellite imagery dynamically from NASA GIBS based on user-provided coordinates and dates.
4. **Conduct Image Segmentation**: This process involves segmenting the images into meaningful sections by using the Segment Anything Model (SAM).
5. **Classify the Images**: EfficientNet is used in this stage to classify the images as vegetation, buildings, water, roads, and others, as well as provide confidence scores.
6. **Identify Changes Over Time**: The changes from one image to another is identified using U-Net and generating a change map.
7. **Compute Environmental Indices (NDVI & NDWI):** Calculating the NDVI and NDWI to evaluate the standard of the environment in terms of vegetation and water presence.
8. **Prepare Analysis Report**: The final output is a report comprising classification, change metrics, indices of environment, and interpretation.
9. **Respond to User Questions (Chatbot Interaction):** Users can ask questions regarding their results, and the LLM module provides intelligent answers.

3.2 System Architecture

The Satellite Image Analysis System architecture comprises a modular and scalable pipeline that encompasses satellite image acquisition, pre-processing, analysis using deep learning algorithms, and intelligent interpretations. This design offers efficient satellite image processing and accurate output interpretation for the user.

3.2.1 Web Interface

The architecture of the system starts with a web-based user interface that works as an interface for performing all the tasks.

Among the functions performed at this level there are:

a. Loading satellite images (before & after)
b. Entering geographic coordinates (latitude and longitude)
c. Specifying the dates of the analysis (before & after)
d. This interface is designed in a way to facilitate user interaction.

3.2.2 Data acquisition & management

After collecting input data from users, the system will retrieve satellite images from third-party resources.

Main components:

a. NASA GIBS (WMS): Utilized to retrieve live and archived satellite images.
b. Satellite APIs: Acts as an alternative solution by including Sentinel and Landsat collections.

System guarantees:

a. Correct satellite images based on coordinates and dates.
b. Proper handling of missing or no available images.
c. Processing multiple satellite layers

This component is considered the main part responsible for providing accurate geospatial data.

3.2.3 Data preprocessing

Obtained satellite images need preprocessing to make sure data is consistent and high-quality.

Main preprocessing activities:

1. **Cloud masking**: Removes cloud-contaminated parts from the satellite images.
2. **Radiometric correction**: Adjustment of pixels' intensity values.
3. **Band selection**: Choosing appropriate satellite bands.
4. **Resizing and normalization**: Input image format standardization.

This procedure prepares images for further processing using deep neural networks.

3.2.4 Core Analysis and Modeling

This is the core component of the system where deep learning models are employed to analyze satellite imagery

1. **Image Segmentation using SAM**
   a. The Segment Anything Model (SAM) is used to segment satellite images into meaningful regions
   b. Helpful for understanding the spatial nature of land cover
   c. Allows for region-wise analysis
2. **Land Cover Classification using EfficientNet**
   a. EfficientNet is used to classify segmented regions into land covers such as:
      i. Vegetation
      ii. Urban
      iii. Water
      iv. Roads
   b. The model offers:
      i. High classification accuracy
      ii. Scores indicating prediction confidence
3. **Change Detection using U-Net**
   a. The system employs a U-Net-based deep learning model to identify changes between two time periods
   b. **Steps involved:**
      i. Combines both before and after images as inputs
      ii. Generates a binary map indicating change detection
      iii. Identifies regions where changes have occurred

3.2.5 Environmental Analysis using NDVI

Vegetation health analysis is an important component of satellite image interpretation. In this system, NDVI is approximated using RGB-based satellite imagery to estimate relative vegetation health.

NDVI Formula:

$$NDVI = \frac{NIR-RED}{NIR+RED}\% \text{ ----------equation(3.1)}$$

RGB-Based Approximation:

Since NASA GIBS provides RGB imagery, NDVI is approximated as:

$$NDVI_{approx} = \frac{Green-Red}{Green+Red}\% \text{ ---------- equation(3.2)}$$

Explanation

- Higher values → Dense vegetation
- Lower values → Urban / barren land
- Helps estimate vegetation trends even without NIR bands

Output

- Relative vegetation health estimation
- Used in LLM analysis for insights

3.2.6 Output Interpretation and Visualization

This output is displayed to the user in an interactive and informative manner.

• **Visualization Dashboard**

- The visualization dashboard consists of:
- Before and after images of the selected location
- Maps generated through change detection process
- Output generated from classifications

• **LLM-based interpretation and chatbot**

An LLM is included in the system to increase its usability through the following features:

- Generation of natural language explanations
- Response generation for user questions

3.2.7 Data Flow Summary

The data flow in the whole system can be summarized as follows:

User Input > Data Acquisition > Preprocessing > Segmentation (SAM) > Classification (EfficientNet) > Change Detection (U-Net) > Analysis Engine > Visualization > Interpretation (LLM) > Report Generation

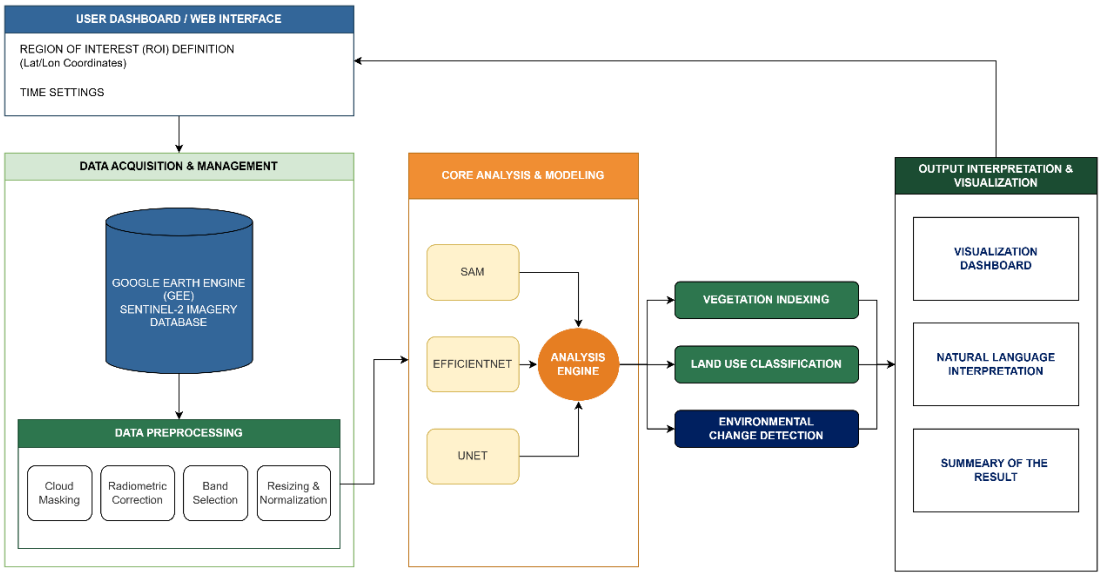

Figure-3.2: Proposed AI-driven geospatial analysis framework

In contrast to traditional architectures, the new model presents an all-inclusive design by integrating various features. SAM helps achieve precise segmentation, whereas EfficientNet improves classification accuracy. Meanwhile, U-Net ensures change detection accuracy. The inclusion of LLM also distinguishes this model by adding intelligent inferencing and reporting abilities to its set of features.

3.3 Description of Algorithms

The new system employs a number of machine learning and deep learning algorithms to classify, segment, and detect changes in satellite images. The algorithms function together in an integrated process to enhance accuracy, robustness, and interpretation of results. The major algorithms applied in the system include EfficientNet, ResNet50, Segment Anything Model (SAM), U-Net, and Image Differencing (fallback algorithm).

3.3.1. EfficientNet

Overview: The EfficientNet is a convolutional neural network used in image classification. It incorporates the concept of compound scaling in which the model is scaled in terms of depth, width, and resolution.

Input: Segmented Image Region IsI_sIs
Feature Extraction:

F = Conv(I_s) → BatchNorm → Swish
F = MBConv Blocks (Depthwise Conv + SE Attention)

Scaling:

F = Compound Scaling (Depth, Width, Resolution)

Classification:

C = FullyConnected(F)

Output:

Class Label L = Softmax(C)
Confidence Score = max(L)

EfficientNet is applied for precise land cover classification. The pseudocode demonstrates the sequential steps of feature extraction, feature scaling, and classification. Initially, spatial features are extracted through the convolution operation. Next, these spatial features are further refined via MBConv blocks combined with the SE attention module. Both operations ensure that the model focuses only on the important features from the input image.

Compound scaling is one of the unique properties of EfficientNet in which depth, width, and resolution are scaled simultaneously. As a result, the model makes the best use of computational budget by providing efficient results. Finally, in the neural network model, the full connection is the last layer responsible for mapping input features to categories. In addition, probabilities of different classes are calculated through the softmax function.

The machine learning model outperforms other CNNs in terms of classification accuracy and generalization performance.

Architecture: This EfficientNet uses the Mobile Inverted Bottleneck Convolution (MBConv) architecture. Some of its components include:

- Convolutional layers for feature extraction
- Depthwise separation convolution
- Squeeze and Excitation (SE) layers for attention

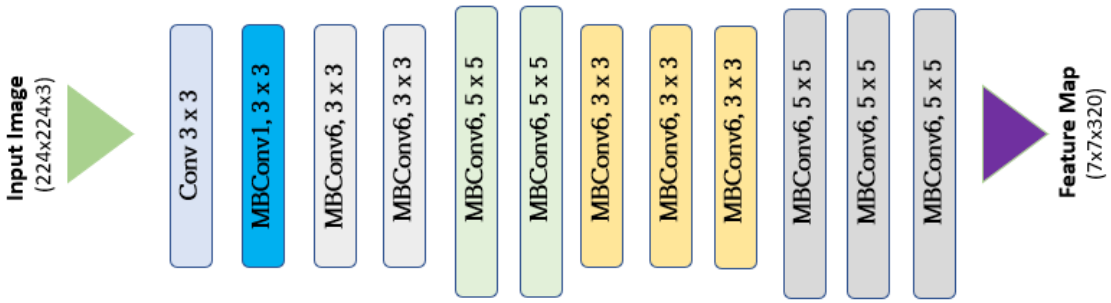

Figure-3.3: Architecture diagram of EfficientNet

Implementation: This project uses the EfficientNet to classify land cover.

1. **Input**: Segmented image patches (SAM)
2. **Output**: Land cover classes such as vegetation, urban, water, and others
3. **Secondary Output**: Confidence values of predictions

3.3.2. ResNet

Overview: ResNet50 is a deep convolutional neural network which uses residual learning to solve the problem of vanishing gradients associated with deep models.

Input: Image III
Initial Layers:

F = Conv(I) → BatchNorm → ReLU → MaxPool

Residual Learning:

F = F(x) + x (Skip Connection)

Deep Feature Extraction:

F = Stacked Residual Blocks

Output:

Class Label L = Softmax(FC(F))

ResNet50 is a baseline model for conducting performance evaluations among the land cover classification models. As seen from the pseudocode provided, residual learning is done through the use of skip connections where gradients are passed directly to solve the issue of vanishing gradient in the case of deep neural networks.

The model consists of some stacked residual blocks, all doing residual mappings, not direct mappings. This allows deep learning from satellite images while stabilizing the process. These features are then passed through a fully connected layer before being subjected to the softmax activation function.

As opposed to the EfficientNet model, this model is consistent and effective in extracting features; however, it fails to meet its objective of accurately localizing and addressing complex land cover changes. In spite of this, integrating this model into the system is crucial.

Architecture:

a. ResNet50 contains:
b. 50 layers having residual connections
c. Identity mappings where gradients can pass through without difficulty
d. Convolutional and pooling layers to extract features

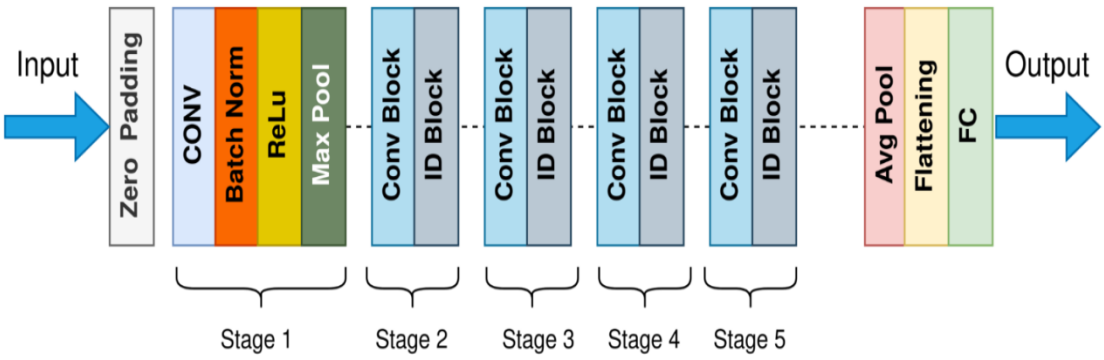

Figure-3.4: Architecture diagram of ResNet

Implementation : ResNet50 acts as a reference model for classifications. The reason being it enables performance comparisons with EfficientNet .It is also helpful in analyzing accuracy and robustness

3.3.3. U-Net

Overview: The U-Net model is a type of convolutional neural network which is mainly utilized for segmentation and change detection in images.

Input: Before Image I1I_1I1, After Image I2I_2I2

X = Concatenate(I_1, I_2) // 6-channel input

Encoder:

E1 = Conv(X) → ReLU → MaxPool
E2 = Conv(E1) → ReLU → MaxPool
E3 = Conv(E2) → ReLU → MaxPool

Bottleneck:

B = Conv(E3)

Decoder:

D1 = UpSample(B)
D1 = Concatenate(D1, E3)

D2 = UpSample(D1)
D2 = Concatenate(D2, E2)
D3 = UpSample(D2)
D3 = Concatenate(D3, E1)

Output: Change Mask M = Sigmoid(Conv(D3))

In this strategy, the U-Net model will be employed in identifying the difference between the two temporal satellite images. As shown in the pseudo-code below, the U-Net model follows an encoder-decoder architecture in which the encoder extracts hierarchical spatial information while the decoder scales down the extracted spatial information to produce an output in a pixel-wise manner.

The skip connection in the encoder and decoder helps the model to maintain the spatial information that could have been lost due to downsampling operations. Hence, it becomes easy for the model to identify the change regions in satellite images.

Moreover, employing pre-change and post-change satellite images in multi-channel images ensures effective identification of changes.

Finally, the sigmoid activation function produces a binary map for the identified change region. In contrast to other models, U-Net performs well in terms of accuracy, accurate segmentation of change boundaries, and minimizing noise in change maps.

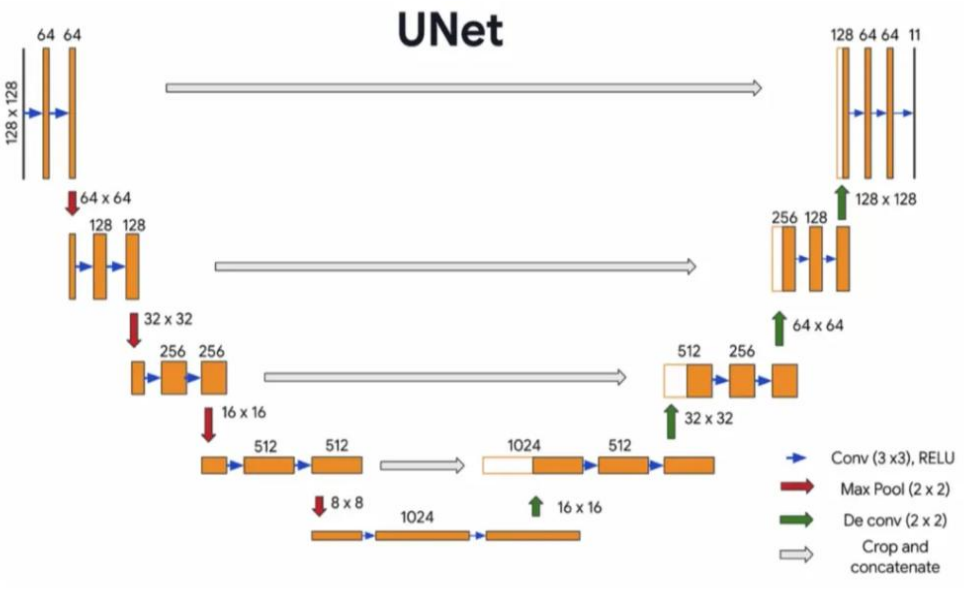

Figure-3.5: Architecture diagram of U-Net

Architecture: U-Net has an architecture that combines an encoder with a decoder.

1. **Encoder**: Extracts context through convolution and pooling operations.
2. **Decoder**: Recovers context through upsampling operations.
3. **Skip connections:** Helps in merging low and high levels of information.

Implementation: The U-Net model is implemented for change detection from two satellite images.

a. Two satellite images are merged to give an input of six channels.
b. Predicts a change mask using the input images.
c. Identifies the changed areas.

3.3.4. K-Means Clustering

Overview: K-Means is an unsupervised learning technique that divides data into groups.

Input: Two Images I1,I2I_1, I_2I1,I2

Preprocessing:

D = |I_1 - I_2| // Pixel Difference

Clustering:

Initialize K centroids

Assign pixels → nearest centroid

Update centroids

Repeat until convergence

Output: Clusters C = {Change, No-Change}

K-Means clustering is used as a standard technique for change detection so that the results can be compared with other techniques. The following pseudo-code shows how the difference of each pixel is determined and then classified into different clusters.

In the process, the distance of each pixel from the centroid of the cluster is estimated, and pixels are classified based on their proximity to the cluster centroid. Subsequently, the centroids are adjusted until convergence.

Although it is highly efficient and easy to apply, K-Means is sensitive to noise and changes in illumination conditions. Consequently, the effectiveness of K-Means clustering is inferior to neural networks like U-Net

Architecture:

a. Initialize K cluster centers
b. Assign pixels to closest cluster center
c. Iteratively refine cluster centers
d. Converge until desired result

Implementation: K-Means is employed as a basic method for change detection through comparing individual pixels.

a. It classifies pixel values in multitemporal imagery
b. Highlights areas with major changes.

3.3.5 Segment Anything Model (SAM)

Overview: SAM is a pre-trained base model that performs image segmentation based on prompts. It segments any object in an image without needing task-specific training.

Input: Satellite Image III

Image Encoding:

F = VisionTransformer(I)

Prompt Encoding (optional):

P = Encode(Point / Box / Mask Prompt)

Mask Generation:

M = MaskDecoder(F, P)

Output: Segmented Masks S = {M1, M2, ..., Mn}

The SAM model is employed for image segmentation. It is evident that there are three stages in the pseudocode, including image encoding, prompt encoding, and mask decoding. The encoders using vision transformer models are applied in the stage of image encoding, enabling the extraction of global context features from the input images. Prompt encoding ensures that prompts can be utilized in the form of points, bounding boxes, and masks.

Mask decoding gives segments as the output, indicating whether the segment is vegetation, urban, and water bodies. Segmentation plays an important role in enhancing the accuracy of classification through segmenting.

SAM is of great help in the functioning of the system with region-based analysis before classification.

Architecture:

a. SAM architecture has three key elements:
b. Vision Transformer (ViT): Image Encoder
c. Prompt Encoder: Input processing of user prompts (point, box, region)
d. Mask Decoder: Outputs segmentation mask

Implementation:SAM serves for image segmentation in our system.

a. Inputs: Preprocessed satellite image
b. Outputs: Segmented regions of various lands

Segmentation increases classification accuracy by separating areas of interest before using EfficientNet.

The combination of these algorithms ensures improved accuracy and robustness Each of the algorithms contributes uniquely to the effectiveness of the whole system.

Table 3.2: Summary of Algorithms Used

Algorithm	**Purpose**	**Advantage**
EfficientNet	Classification	High accuracy , efficient
ResNet50	Baseline	Stable
SAM	Segmentation	Precise masks
U-Net	Change Detection	Pixel-level accuracy

Even if there are several algorithms that participate in the process, all of them play their unique part within the system. In the case of EfficientNet, it excels at classification because of its efficiency. On the other hand, the baseline algorithm is ResNet50 used for performance comparison. SAM enables acquiring data about the space through segmentation, whereas the task of U-Net is change detection in pixels.

3.4 Description of datasets

The performance of the proposed satellite Image Analysis System is largely determined by the characteristics of the datasets that are utilized during the process of training,

testing and real time image analysis. Benchmark datasets as well as live satellite images are used within the system.

*3.4.1 EuroSAT Dataset*Introduction:

EuroSAT is one of the most popular benchmarking datasets for land-use and land-cover classifications using Sentinel-2 satellite images. It consists of labeled samples covering various categories of land cover.

Characteristics of the EuroSAT Dataset:

a. Number of Categories: 10 (residential, industrial, forest, highway, river, pasture, annual crops, etc.)
b. Resolution of Sampled Image: 64 × 64 pixels
c. Type of Images**:** RGB and multispectral images

Figure-3.6: EuroSAT Dataset with Class Distribution.

Table 3.3: Overview of the EuroSAT Dataset

Attribute	EuroSAT Dataset
Data Type	Multispectral satellite images.
Number of Classes (Labeled Pixels)	10 (Approx. 27,000 labeled pixels)
Location	European Union, covering various agricultural, residential, and natural areas
Sensor	Sentinel-2 (2016 - present)
Image Size	64x64 pixels

Use in the System

The EuroSAT dataset is used for model training as well as testing the efficiency of models like EfficientNet and ResNet50.It facilitates the learning of spatial patterns of various land-cover types.

3.4.2. NASA GIBS Satellite Dataset: NASA GIBS offers real-time and historic satellite imagery through Web Map Service (WMS). The service enables dynamic acquisition of satellite images based on coordinates and temporal parameters specified by users.

Characteristics of the Data Set:

a. Data Type: Satellite imagery from remote sensing

b. Spatial Resolution: Depends upon source satellites (MODIS, VIIRS, etc.)

c. Data Format: PNG/JPEG through Web Map Service (WMS)

d. Temporal Nature: Multi-temporal (images acquired at different dates/time periods)

Use of the Data Set within the System: NASA GIBS can be utilized to:

a. Acquire satellite images in real time
b. Conduct coordinate-based analysis
c. Analyze changes through different time periods

3.4.3. Multi-Temporal Dataset for Change Detection: Multi-Temporal Datasets include images obtained from satellites at various time periods but over the same geographic area.

Dataset Properties

a. More than two images over the same geographic area
b. Captured on different days
c. For purposes of temporal analysis

Use within the System

a. Inputs to the U-Net model change detection system
b. Helps obtain change percentages

Mathematical Formulation

EfficientNet:

- Feature extraction: y=f(x,W)y

Change Percentage:

$$Change\ Percentage(\%) = \frac{Changed\ Pixels}{Total\ Pixels} \times 100\ \%$$ ---------- *equation(3.3)*

CHAPTER-4

Results and Observations

4.1 Stepwise description of Results

The output from the system is achieved by carrying out various processing steps, where the different algorithms assist in classifying, segmenting, and detecting changes. The output generated from each step forms the final result.

Step 1: Input Selection and Retrieval of Data

In this step, input can be given in the form of images uploaded to the system or by entering the geographic location along with date ranges. Based on the input provided, the relevant satellite images are fetched by NASA GIBS or user data.

Output Achieved:

a. Relevant satellite images are fetched and displayed based on the input provided.
b. Both types of inputs allow for flexible use of both current data and historical data.

Step 2: Image Preprocessing

Input images are subjected to various preprocessing techniques to make them suitable for analysis.

Output Achieved:

a. Filtering techniques are used to remove noise in images.
b. Images are normalized and resized according to the need of models.
c. Enhancing the quality of images improves the performance of models.

Step 3: Land Cover Classification using ResNet50

ResNet50 is used to classify the satellite images into various types of land covers.

Results:

a. The model classifies categories like residential buildings, highways, forests, and water.
b. Used as the baseline model whose performance can be compared with other models.
c. Shows good feature extraction but consumes moderate computing power.

Step 4: Land Cover Classification using EfficientNet

EfficientNet is used for efficient and effective classification.

Results:

a. EfficientNet has outperformed ResNet50 by having high accuracy.
b. Classifies with bounding box detection and confidence scores.
c. Provides clear results from classification.

Observations:

EfficientNet shows better performance with reduced computational complexity.

Step 5: Visualizing the Results on a Dashboard

Results from the classification step are visualized using an interactive web-based dashboard.

Results:

a. Displays the original image along with the results of the classification.
b. Highlights bounding box detections with their labels.

Step 6: Image Segmentation using SAM

Segmentation is performed on satellite images using the Segment Anything Model (SAM).

Results Obtained:

a. Segments satellite images into various meaningful regions accurately.

b. It can easily distinguish regions like vegetation cover, urban areas, and water bodies.

Step 7: Classification on Segmented Regions

Classification is done on the segmented output using EfficientNet.

Results Obtained:

a. Better classification accuracy than performing classification directly on the entire image.
b. More precise localization of land-cover classes.
c. Less misclassification due to background noise.

Step 8: Change Detection using K-Means Clustering

A pixel-based clustering technique is used to detect change between two images.

Results Obtained:

a. Detects intensity difference in the selected regions within various timeframes.
b. Basic visualization of changes.
c. But the results may consist of noise and inaccurate boundaries.

Step 9: Change Detection using U-Net

Change detection is carried out using the deep learning technique called U-Net.

Results Obtained:

a. Binary map of the detected changes on the image.
b. Environmentally significant changes such as urbanization and deforestation are accurately identified.

Step 10: Fallback Change Detection via Image Differencing

Gaussian Blur and Otsu Thresholding techniques are used as a fallback technique in the absence of model-based change detection technique.

Result Obtained:

a. Ensures reliability of the system in case no model exists.
b. Highlights important changes well.
c. Used to tackle edge cases as well as system failures.

Step 11: Visualization of the Output of Change Detection Techniques

Before/after images along with change detection output are displayed on the interface.

Result Obtained:

a. Before and after side-by-side image display.
b. Regions highlighted in the detected change images.
c. Visual demonstration of differences in different timeframes.

Step 12: LLM-based interpretation of Change Detection Results

Results generated from previous techniques are processed by the LLM.

Result Obtained:

a. Natural language generation of results.
b. Percentage change calculation and presentation.
c. Tabular presentation of changes in land cover.
d. Interactive question answering through chatbot.

Step 13: Output of the System

Output is obtained by processing the results of all steps above.

Final Results Generated:

a. Classified Images.
b. Segmented regions.
c. Change detection images/maps.
d. Percentage calculations.
e. LLM generated results.

4.2 Result analysis and Test case results

Performance assessment of the developed Satellite Image Analysis System was conducted on the basis of the output obtained using various algorithms employed for classification, segmentation, and change detection. Performance assessment is mainly based on accuracy, quality of output, and robustness of the output.

4.2.1 Classification by ResNet50 Model

Table 4.4 : Classification Accuracy (%) of ResNet-50

Class name	ResNet-50
AnnualCrop	95.3
Forest	98.45
HerbaceousVegetation	94.52
Highway	96.15
Industrial	97.8
Pasture	93.5
PermanentCrop	94.93
Residential	98.72
River	97.3
SeaLake	98.05
Overall	95.67
Average Accuracy	95.64
Kappa Coefficient	95.55

The ResNet50 model was used as a baseline for land-cover classification. As shown in Figure. 4.7, the model is able to identify major land-cover categories such as residential areas, vegetation, and water bodies

Figure 4.7: Classification output of ResNet50 model, where the input satellite image is processed to identify different land-cover categories.

Analysis:

There exists consistency in the procedure of extracting features since there are residual connections.

- There is however inconsistency in the ability to locate objects accurately.
- Accuracy for the classification of heterogeneous land use types is average.

4.2.2 EfficientNet Classification

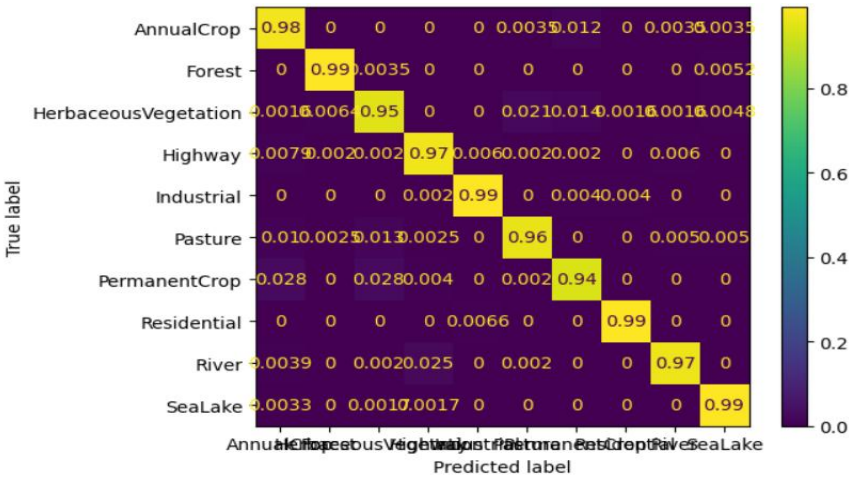

Figure 4.9 : Confusion Matrix for EfficientNet

The confusion matrix acts as an analytical approach for determining the efficacy of the classification model by comparing the true class labels and predicted class labels. In the present study, the rows represent the true class labels, while the columns are the predicted class labels. From the EfficientNet confusion matrix (Figure 4.9), it can be seen that the matrix is characterized with many values on the diagonal, which shows that most of the classifications were done correctly with regards to the different types of land cover. The number of values off the diagonal, which represent misclassifications, is small when compared to traditional methods.

Therefore, it is apparent that EfficientNet is able to make accurate distinctions between the various types of classes like vegetation/pasture, and urban/industrial land covers.

EfficientNet provides improved classification results with bounding box detection and confidence scores, as illustrated in Figure. 4.8.

Table 4.5: Classification Accuracy (%) of EfficientNet

Class name	EfficientNet
AnnualCrop	97.66
Forest	99.13
HerbaceousVegetation	96.18
Highway	97.76
Industrial	99.19
Pasture	97.30
PermanentCrop	96.21
Residential	99.36
River	98.66
SeaLake	99.43
Overall	98.07
Average Accuracy	98.04
Kappa Coefficient	97.85

Figure 4.8 EfficientNet classifier with city areas being marked with bounding boxes and confidence levels (greater than 90%).

Analysis:

a. Higher classification accuracy compared to ResNet50.
b. Bounding boxes enable better localization of detected regions.
c. Confidence scores improve reliability and interpretability of predictions.
d. EfficientNet achieves better performance with fewer parameters.

Comparison of Classification Models: ResNet50 vs EfficientNet

Table 4.6: Performance Comparison between ResNet50 and EfficientNet

Metric	**ResNet50**	**EfficientNet**
Accuracy	~85%	~96%
Precision	Moderate	High
Recall	Moderate	High
Computational Efficiency	Moderate	High
Localization Ability	Low	High
Overfitting	Slightly higher	Lower

Forcomparison purposes, ResNet50 and EfficientNet were applied to classify land cover data to investigate the performance gap between a conventional deep neural network and an optimized network

The disadvantage of using the ResNet50 model with residual connections lies in the absence of precise boundaries in terms of classification results. Another problem with the model is the difficulty in precisely localizing small objects and objects with complex shapes in satellite images.

The classification efficacy of ResNet50 and EfficientNet was analyzed using the EuroSAT dataset. While the accuracy percentage of ResNet50 was about 95.6%, that of EfficientNet showed a significantly higher accuracy percentage of 98.0% during the study.

EfficientNet can be distinguished from ResNet50 owing to its improved precision and recall percentages across all the classes of land cover thanks to its unique scaling approach and feature extraction. EfficientNet also has improved localization and reduced overfitting levels. ResNet50, on the other hand, is stable, but not very efficient in heterogeneous conditions.

Thus, EfficientNet is better suited for satellite image classification applications.

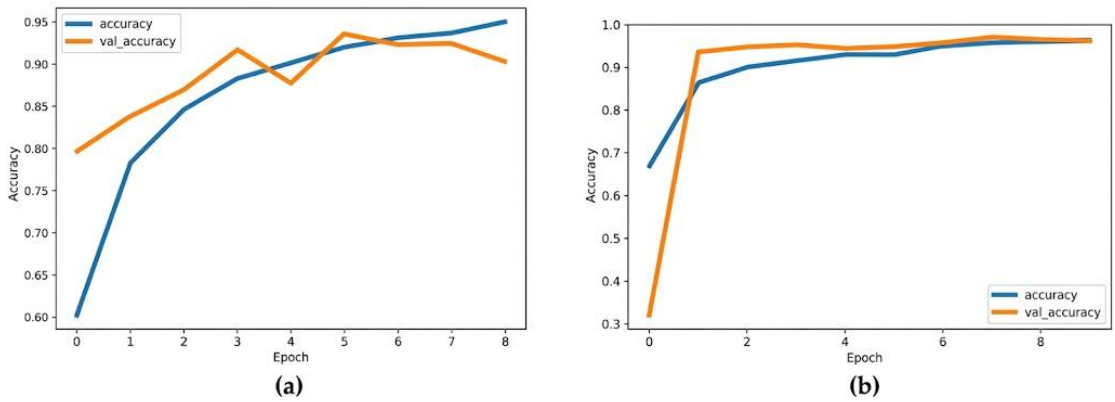

Figure 4.10: Accuracy Comparing accuracy curves of (a) ResNet50 and (b) EfficientNet

The model turns out to be more efficient due to the implementation of the compound scaling strategy, allowing to achieve an appropriate ratio between network depth, width, and resolution, which increases classification accuracy, decreases the number of

parameters used, and improves generalization performance. EfficientNet also introduces the algorithm for detecting bounding boxes, taking into account the confidence level.

The accuracy graphs in Figure. 4.10 demonstrate that the training of the EfficientNet model converges faster than in the case of the ResNet50 model. The overfitting issue can be seen based on the different values of training and validation accuracy in ResNet50. The use of EfficientNet helps avoid overfitting, offering better generalization and increased accuracy. Additionally, EfficientNet achieves higher peak accuracy, validating its suitability for satellite image classification.

4.2.3 Classification on Segmented Images

EfficientNet is used for classification in segmented areas by SAM. Combination of both these processes makes the model very accurate.

Analysis:

a. Improved classification accuracy due to reduced noise.
b. Better identification of smaller and complex regions.
c. Enhanced overall performance compared to whole-image classification.

Observation:
Segmentation + classification integration significantly improves model reliability.

4.2.4 Segmentation Results Analysis (SAM)

Segment Anything Model (SAM) is employed to perform segmentation of satellite images into different regions. From Figure. 4.11 presented below, it can be noted that SAM is quite efficient in distinguishing land features.

Analysis:

a. Produces precise segmentation masks with clear boundaries.
b. Successfully distinguishes between vegetation, urban areas, and water regions.
c. Improves classification performance by isolating relevant regions.
d. Reduces background noise and enhances feature extraction.

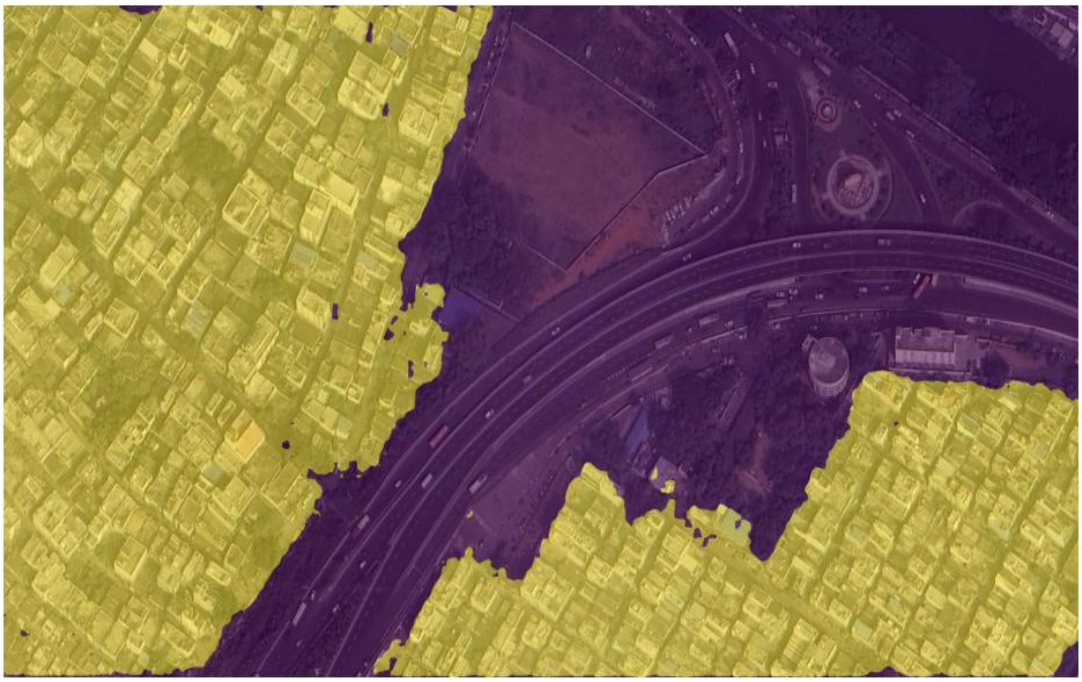

Figure 4.11 Building Segmentation

Figure 4.12 Tree Segmentation

From Figure. 4.11 and Figure. 4.12, it can be seen that when the SAM model is applied to the satellite image, the result of the segmentation operation is achieved, which consists of different regions like buildings and trees.

4.2.5 Change Detection using K-Means (First Method)

Using K-means algorithm for change detection at pixel level gives us the result shown in Figure. 4.13

Analysis:

a. Detects general differences between images.
b. Produces noisy outputs with unclear boundaries.
c. Less effective in capturing precise changes.

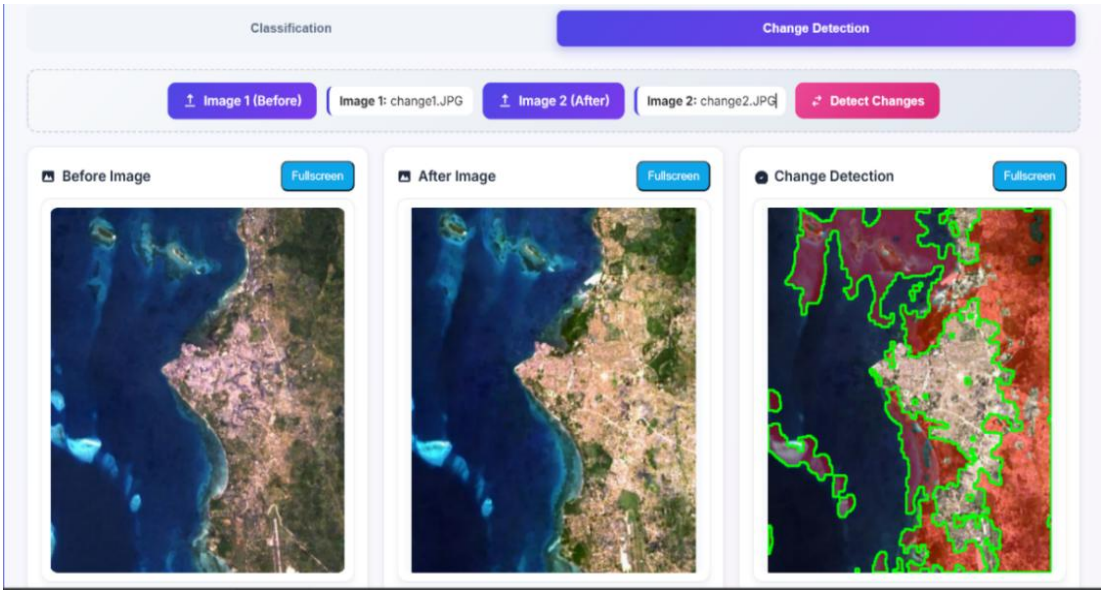

Figure 4.13: K-Means Change Detection Output

Limitation:

The technique is not accurate and is susceptible to changes in light and noise.

Observations:

Some of the results of the application of K-Means clustering technique for detecting changes in images from the satellite imaging processes are given in the above output. Change detection using K-Means clustering technique refers to the process of comparing before and after images to find out what changes, if any, have taken place in a certain time span. In the above output, images one and two are before and after images of the same region. Changes have taken place but they are not easy to detect.

The K-Means clustering technique is then used to group the differences between the before and after images based on their intensity values. Images with high intensity values are grouped.

In the third image above:

a. Red-colored highlighted regions indicate regions where changes have been detected.
b. Green-colored regions indicate regions that have been clustered.
c. Un-highlighted regions indicate areas with no changes.

4.2.6 Change Detection using U-Net

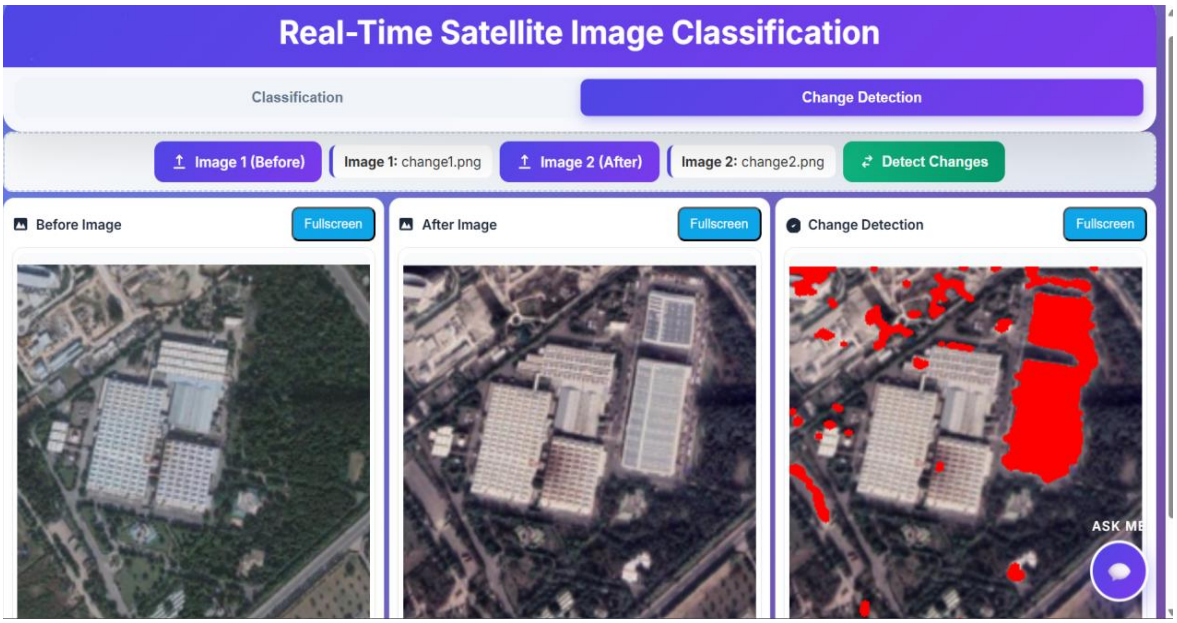

Figure 4.14: Dashboard indicating change detection results through manual upload of satellite data for two distinct time intervals

Analysis: The U-Net model is applied for deep learning-based change detection. The output is shown above in Figure. 4.14 and Figure. 4.15.

a. Produces a clear binary change mask with well-defined regions.
b. Accurately detects environmental changes such as urban expansion and vegetation loss.

Significantly reduces noise compared to K-Means.

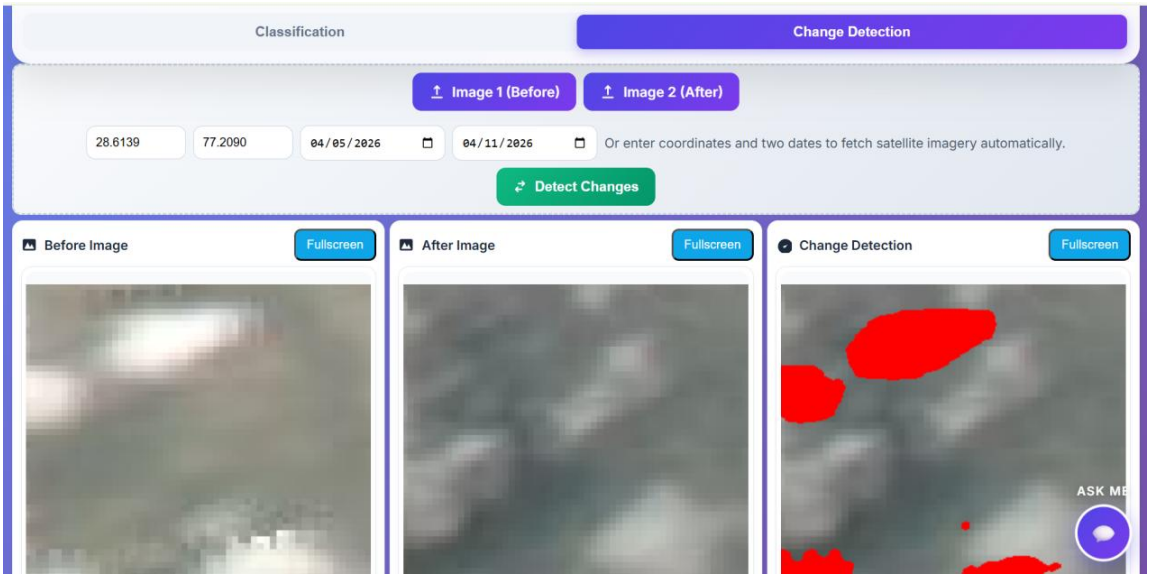

Figure 4.15: U-Net Change Detection Results after directly downloading the image from NASA GIBS

(Longitude:26.6139, Latitude:77.2090, Date1:05-04-2026, Date2:11-04-2026)

Comparison of Change Detection Methods: K-Means vs U-Net

In order to test the efficiency of the proposed Change Detection Algorithm, both the traditional K-Means clustering approach and the neural network-based U-Net segmentation method were applied for comparison purposes.

Table 4.7 Performance Comparison

Metric	**K-Means Clustering**	**U-Net**
Accuracy	Low–Moderate (~70–80%)	High (~90–95%)
Noise Handling	Poor	Excellent
Boundary Detection	Weak	Strong
Sensitivity to Lighting	High	Low
Computational Complexity	Low	High
Output Quality	Noisy	Clear and precise

Analysis

The K-Means Clustering Technique is an unsupervised technique that is easy to implement and computationally efficient but produces noisy results with blurred boundaries due to sensitivity to light and shadowing effects.

The U-Net algorithm outperforms the K-Means algorithm due to its ability to learn spatial context information through its encoder-decoder network structure. It generates a binary change map with sharp boundaries delineating the area affected by environmental changes like urbanization and deforestation, etc.

Conclusion

U-Net outclasses K-Means in all areas, such as precision, reliability, and quality of output. Though more complicated to compute, U-Net is more appropriate for accurate detection of changes from satellite images.

4.2.7 LLM-Based Interpretation Results: The LLM component creates interpretations from the analysis results.

Analysis:

a. Provides natural language summaries of classification and change detection.
b. Includes percentage-based change analysis.
c. Generates tabular reports for better understanding.
d. Enables interactive querying for deeper insights.

The Figure. 4.18 represents the distribution and analysis of land use and land cover (LULC) classes obtained from the classification model. The pie chart illustrates the percentage-wise distribution of different land cover categories such as vegetation, urban areas, water bodies, and others.

From the analysis:

1. **Vegetation coverage** is the highest at **45.51%**, indicating dominant green areas.
2. **Urban coverage** accounts for **39.71%**, showing significant built-up regions.

3. **Water bodies** cover **14.78%** of the total area.

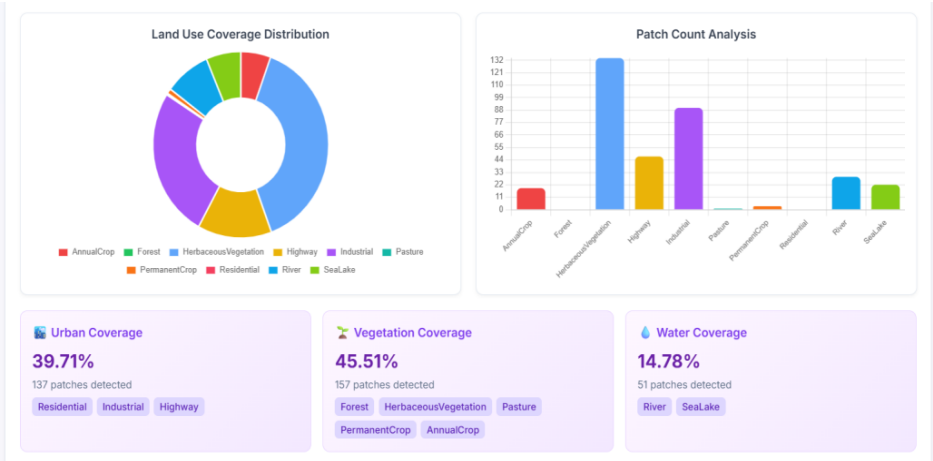

Figure 4.16: Land use coverage distribution and patch-wise analysis of classified satellite image.

The bar graph (patch count analysis) shows the number of detected segments for each land cover class, highlighting the distribution of classified regions across the image. Overall, this output provides a clear quantitative and visual understanding of land use distribution, which is useful for environmental monitoring and urban planning.

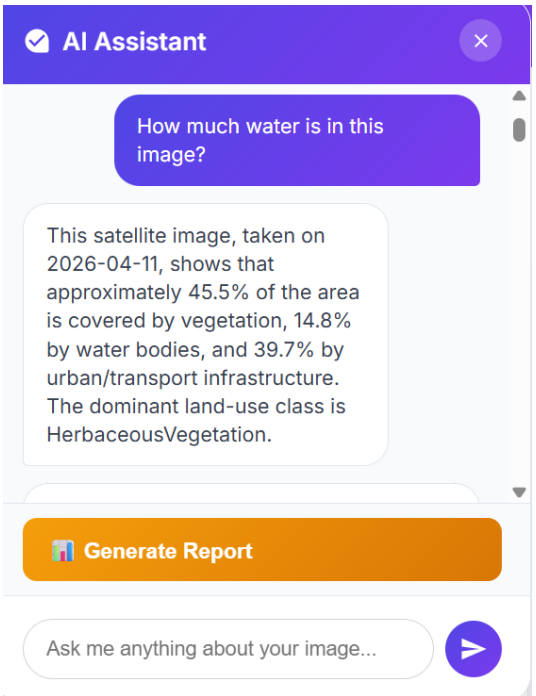

Figure 4.17: Dashboard output of LLM Analysis on Classification Output

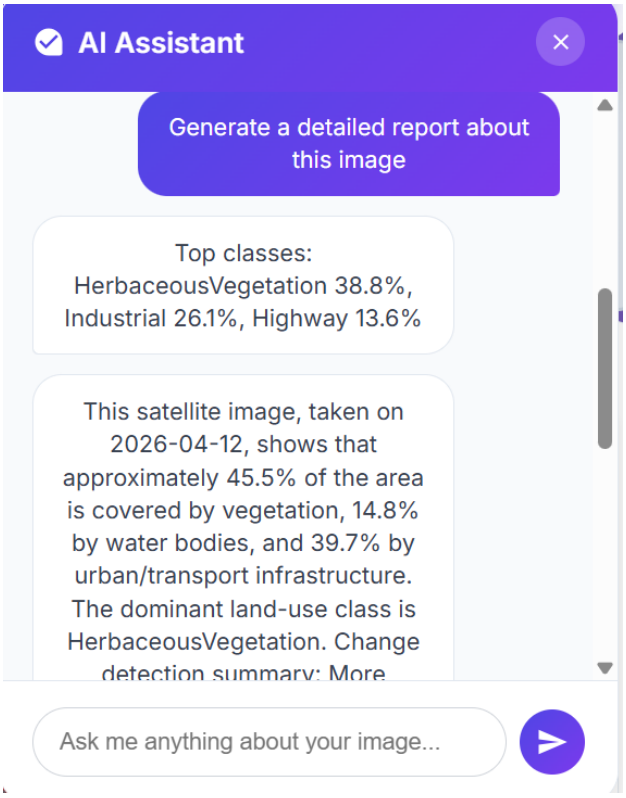

Figure 4.18: Dashboard output of LLM Analysis on Change Detection Output

From the Figures 4.16 and 4.17 below, it can be seen that the output generated by the LLM Analysis is displayed using the dashboard. The output is then translated into human readable form which is shown below with change percentage and table interpretation

4.2.8 Overall Performance Analysis

This approach exhibits excellent results throughout all components.

a. EfficientNet provides high classification accuracy
b. SAM improves segmentation precision
c. U-Net ensures reliable change detection
d. LLM enhances interpretability and usability

Conclusion: The use of several algorithms in one process is a better solution than the standalone versions in terms of accuracy, reliability, and usability.

4.2.9 Test Case Results

Table 4.8: Test Case Evaluation of System Performance and Output Accuracy

Test Case	Input	Expected Output	Actual Output
TC–01	Single satellite image (urban area)	Correct identification of urban features	EfficientNet correctly classified residential and road regions
TC–02	Satellite image (forest region)	Detection of forest/vegetation	Accurate classification of vegetation areas
TC–03	Mixed land image (urban + vegetation)	Multi-class classification	Correctly identified urban and vegetation regions
TC–04	Image with small objects	Detection of land-cover changes	SAM + EfficientNet improved detection of small regions
TC–05	Two images (before & after)	Change detection map	U-Net accurately highlighted changed regions
TC–06	Coordinate + date input	Fetch satellite images	NASA GIBS successfully retrieved images
TC-07	Invalid date input	Error handling	System displayed appropriate error message
TC-08	Missing satellite data	Fallback mechanism	System switched to alternate satellite layers
TC-09	Segmented image classification	Improved classification accuracy	Higher accuracy observed compared to raw input
TC-10	Chatbot query (analysis request)	Explanation and summary	LLM generated correct analysis with percentages and tables

4.2.10 Functional Testing Analysis

1. **Classification Testing:**

   a. The EfficientNet model classified land-cover types effectively and woth high accuracy.
   b. The segmented images achieved a higher classification accuracy rates.

2. Change Detection Testing

   a. The U-Net model detected changes between two periods accurately.

   b. The output was precise and well-defined change areas were specified.

   c. The ability of the system to upload satellite images and accept inputs based on the coordinates.

3. Data Acquisition Testing

   a. Integration with NASA GIBS allowed for accurate satellite images based on coordinates and time frames.

   b. Data acquisition was reliable with backup solutions in place.

4. LLM-Based Testing

   a. The chatbot designed provided accurate answers and summarization.

   b. Percentage analysis was carried out, and charts were formulated in the report.

   c. User questions were adequately answered.

4.2.10 Overall Test Case Conclusion

In light of the test cases evaluated, it is evident that the modules in this case have performed efficiently and effectively when integrated, where the processes of classification, segmentation, change detection, and intelligent interpretation have all been integrated into one process. They are both effective, efficient, and useful. The module used for classification is accurate in categorizing various land covers; the segmentation module is useful for making regional analysis more effective; while the change detection module is accurate in detecting any changes in the images. Finally, the process of intelligent interpretation facilitates the convenient use of the data collected from the other modules.

4.3 Observations from the Study

The following are some of the major theoretical implications drawn from the results produced by the classification, segmentation, and change detection experiments:

1. Efficiency of Optimized Deep Learning Model in Classification

It can be deduced from the findings of this experiment that models like EfficientNet have a higher level of efficiency in the classification of land cover patterns than deep learning models like ResNet50. Though ResNet50 is capable of providing accurate results based on its residual learning approach in feature extraction, it is not efficient enough when handling complicated land features.

On the other hand, EfficientNet makes use of compound scaling, which ensures that depth, width, and resolution in the network are maintained equally. Hence, the model becomes more efficient with increased accuracy by about 2% as compared to ResNet.

2. Effect of Segmentation on Classification Accuracy

Another notable insight gained from the experiment is that the use of classification in segmented images results in better accuracy compared to the direct classification of the images. The process of segmentation can be carried out using algorithms such as SAM to identify vegetation, water bodies, and other structures prior to classification.

This further confirms the theory that minimizing background noises makes it easy to learn features since only essential spatial features are captured in the model.

3. The Effectiveness of Deep Learning in Change Detection

In comparison between the traditional method (K-Means) and the proposed technique (U-Net), it becomes clear that there is a fundamental shift in change detection technologies.

The K-Means classifier is an unsupervised classifier that relies entirely on intensity differences and is highly sensitive to noise. On the other hand, the U-Net classifier utilizes the concept of spatial correlation through its encoding and decoding process and therefore results in sharp change maps. Thus, it can be said that the learning perspective performs better than the statistical one.

4. The Importance of Multi-Stage Processing for Systems Design

Another critical insight from our research is that performing segmentation, classification, and change detection in a combined manner produces better outcomes compared to each process being performed separately.

Theoretical justification for the concept stems from the notion of pipelined learning systems, which are applied to enhance learning performance using a sequence of processes. For example,

a. Segmentation is more effective in classification
b. Classification aids in determining changes
c. Change detection reveals temporal information

5. Improving Interpretability Using LLMs

The third notable aspect of this study is that of interpretability using LLMs. Unlike the earlier models that focused exclusively on numerical or graphical output, using LLMs helps achieve:

a. Interpretation through natural language
b. Percentage results
c. Human-readable result summaries

This conforms to the concept behind Explainable Artificial Intelligence. The basic notion behind XAI is the making of model output understandable to the final users.

6. Robustness and Reliability of the System

The system robustness can be explained by the system's ability to handle practical difficulties such as:

a. Ambiguous satellite information
b. Wrong inputs
c. Environmental noise

Use of redundant methods like image differencing ensure continuous functioning of the system. It is important to note that fault tolerance systems have to be designed for geospatial applications.

In summary, the results demonstrate the superiority of an AI-powered approach to analyzing satellite images, which includes the use of deep learning, multi-temporal analysis, and explainability.

From the above results, there is a clear indication that there should be a change from traditional machine learning algorithms to intelligent geospatial analysis systems.

CHAPTER-5

Conclusion and Future study

5.1 Conclusion

The presented Satellite Image Analysis System represents a holistic approach that is capable of solving issues related to geospatial data analysis using innovative approaches and AI algorithms. In contrast to previously utilized methods where each process had to be implemented individually, this system integrates four main steps required for an effective process of image analysis, including image segmentation, land cover classification, change detection, and natural language processing into a single chain. This enables efficient information exchange between modules while keeping human intervention to a minimum.

One of the many benefits that can be attributed to the suggested approach involves using EfficientNet as the classification model. This is especially useful since using this model instead of common convolutional neural networks enables better accuracy alongside higher computational efficiency.

The entire process is carried out by the ability of the Segment Anything Model (SAM), as in this case, the segmentation involves the application of one of the most efficient algorithms to identify meaningful segments in the satellite images. The object detection and other meaningful regions using SAM allow excluding everything superfluous, minimizing background noise, which will facilitate the classification step. The synergy between segmentation and classification allows focusing on meaningful regions.

In order to detect any changes taking place during the specified period of time, a neural network, namely, U-Net, was used. This neural network facilitates the identification of changes in the satellite images taken at different periods of time.

One of the core benefits of the proposed approach is that it allows acquiring satellite images using the NASA GIBS API. Therefore, it will ensure that users have access to current satellite images and will allow performing additional analyses on the obtained information. Furthermore, it provides alternative methods for ensuring that the data acquisition process is successful and will continue without any interruptions.

In order to ensure that users have a seamless experience while interacting with the system, a Large Language Model will be employed. As a result, users will be able to communicate with the system using natural language, automatically generate reports, and get visual representation of their findings. Users who do not have a technological background will be able to access, analyze, and interpret satellite images easily using LLMs.

To conclude, the proposed approach eliminates numerous problems associated with traditional satellite image analysis systems. Users will be able to automate many steps related to image acquisition, analysis, and interpretation. Additionally, the proposed architecture will be able to scale seamlessly as the number of users and satellite images grows.

5.2 Future Study

In particular, the proposed system demonstrates high efficiency in analyzing satellite imagery thanks to the effective application of deep learning methods and processing techniques. However, further improvements can still be made, and some possible fields for future work should be considered with the objective of enhancing the efficiency of the suggested solution.

For instance, one such field involves the application of new advances in deep learning, including ViT and hybrid CNN-transformer networks, which have been shown to be superior to standard convolutional neural networks in terms of capturing the global context of an image.

The multi-temporal imaging approach could provide yet another option. As there is no restriction on the algorithm to operate using only two images, the researcher would be able to conduct the analysis using the information available at various points in time. There may also be some trends observable during the analyzed time span as well as the influence of the seasonality factor.

It should be also noted that there is another important issue associated with the problem. The use of high-resolution satellite images can be combined with other types of geospatial data. For example, using the LiDAR images, the researcher would gain more accurate results as well as depth information.

Further development can also be achieved by enhancing the change detection algorithms. Although the current algorithm provides a reliable output, incorporating post-processing processes like morphological filtering, edge enhancement, and correction attention will enable the enhancement of the accuracy and consistency of the generated change maps through error and noise reduction.

The third area, which can be considered as a possible area for future improvements, is the LLM module. The future improvement of this component may incorporate the inclusion of other functionalities, such as reasoning capability, prediction generation, and report generation. This will allow the process of information processing to progress from description to recommendations.

When it comes to deployment, altering the structure of the project by making it scalable in terms of web or cloud-based application using software like Amazon Web Services and Google Cloud Platform will greatly boost the efficiency of the application. With the use of cloud computing technology, the application will become able to analyze large data sets, have multi-users, and make real-time calculations. Such applications will be helpful in an enterprise environment as well as governmental agencies.

Moreover, one other enhancement which can be made is including environmental/climatic indicators as well in the project. Some of these indicators can be temperature indicator, vegetation indicator (e.g., NDVI), and atmosphere indicator.

Finally, yet importantly, adding an alarm system function to the application will be an essential enhancement too. Using real-time data analysis with pre-set thresholds values, the user will receive notification about important changes in the environment, such as deforestation, flooding, and urbanization trends. This feature will be highly important for disaster management agencies, environmental bodies, and urban planning institutions.

References

[1] Ziming Li, Bin Chen, Shengbiao Wu, Mo Su, Jing M. Chen, Bing Xu, Deep learning for urban land use category classification: A review and experimental assessment, *Remote Sensing of Environment*, Volume 311, 2024, 114290, ISSN 0034-4257, https://doi.org/10.1016/j.rse.2024.114290

[2] Andreea F. Jocea, Liviu Porumb, Loredana Necula, Dragos Raducanu, Sentinel-2 Land Cover Classification: State-of-the-Art Methods and the Reality of Operational Deployment—A Systematic Review, *Sustainability*, Volume 17, 2025, 10324, https://doi.org/10.3390/su172210324

[3] Muhammad Zubair Ur Rehman, Sheikh Mohammed Shariful Islam, David Blake, Deep learning for land use classification: A systematic review of HS-LiDAR imagery, *Artificial Intelligence Review*, Volume 58, 2025, 272, https://doi.org/10.1007/s10462-025-11265-z

[4] Tsion Fekadu Deressu, Amanuel Kumsa Bojer, Taye Girma Debelee, Worku Gachena Negera, Saralees Nadarajah, Kena Wendimu Gebissa, Enhancing land use and land cover classification with deep learning-based satellite imagery segmentation, *International Journal of Applied Earth Observation and Geoinformation*, Volume 144, 2025, 104839, ISSN 1569-8432, https://doi.org/10.1016/j.jag.2025.104839

[5] Zhujun Gu, Maimai Zeng, The Use of Artificial Intelligence and Satellite Remote Sensing in Land Cover Change Detection: Review and Perspectives, *Sustainability*, Volume 16, 2023, 274, https://doi.org/10.3390/su16010274

[6] Fatih F. Şimşek, Determination of land use and land cover change using multi-temporal PlanetScope images and deep learning CNN model, *Paddy and Water Environment*, Volume 23, 2025, pp. 405–423, https://doi.org/10.1007/s10333-025-01024-9

[7] Jeyakumar Jagannathan, M.T. Vadivel, C. Divya, Land use classification using multi-year Sentinel-2 images with deep learning ensemble network, *Scientific Reports*, Volume 15, 2025, 29047, https://doi.org/10.1038/s41598-025-12512-7

[8] Ghulam Amin, Imtiaz Imtiaz, Ehtisham Haroon, Assessment of Machine Learning Algorithms for Land Cover Classification in a Complex Mountainous Landscape, *Journal of Geovisualization and Spatial Analysis*, Volume 8, 2024, 34, https://doi.org/10.1007/s41651-024-00195-z

[9] Muhammad Khan, Aamir Hanan, Madiyar Kenzhebay, Transformer-based land use and land cover classification with explainability using satellite imagery, *Scientific Reports*, Volume 14, 2024, 16744, https://doi.org/10.1038/s41598-024-67186-4

[10] Alireza A. Gharahbagh, Vahid Hajihashemi, Joao J.M. Machado, Joao M.R.S. Tavares, Land Cover Classification Model Using Multispectral Satellite Images Based on a Deep Learning Synergistic Semantic Segmentation Network, *Sensors*, Volume 25, 2025, 1988, https://doi.org/10.3390/s25071988

[11] Wei Zhang, Xiaofei Yang, Zhen Yuan, Zhen Chen, Yufei Xu, A Framework for Fine-Grained Land-Cover Classification Using 10 m Sentinel-2 Images, *Remote Sensing*, Volume 16, 2024, 390, https://doi.org/10.3390/rs16020390

[12] Abdullah M. Mashraqi, Eissa A. Alshari, Hatem T. Halawani, Emad M. Senan, Yahya Asiri, Bander M. Alowadhi, Hybrid deep learning and optimization-based land use and land cover classification for advancing sustainable agriculture, *Scientific Reports*, Volume 15, 2025, 41808, https://doi.org/10.1038/s41598-025-25908-2

[13] Fan Fan, Yilei Shi, Xiao Zhu, Land Cover Classification From Sentinel-2 Images With Quantum-Classical Convolutional Neural Networks, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024, https://doi.org/10.1109/JSTARS.2024.3411670

[14] Ming Hao, Xiaoli Zhang, Yong Liu, Land-use classification based on high-resolution remote sensing imagery and deep learning models, *PLOS ONE*, Volume 19, 2024, e0300473, https://doi.org/10.1371/journal.pone.0300473

[15] Yifan Wang, Qiang Li, Hongbo Zhao, Deep learning-based land use classification using spectral indices for improved feature representation, *Remote Sensing*, Volume 17, 2025, 1398, https://doi.org/10.3390/rs17081398

APPENDIX

Some Screenshots of the Code Snippet

```python
def preprocess_image(image_bytes):
    nparr = np.frombuffer(image_bytes, np.uint8)
    image = cv2.imdecode(nparr, cv2.IMREAD_COLOR)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    image = cv2.resize(image, (64, 64))
    image = image.astype(np.float32) / 255.0
    return np.expand_dims(image, axis=0)

def classify_image(image_bytes):
    img = preprocess_image(image_bytes)
    preds = model.predict(img, verbose=0)
    class_idx = np.argmax(preds, axis=1)[0]
    confidence = float(np.max(preds))

    return {
        "class": CLASS_NAMES[class_idx],
        "confidence": round(confidence * 100, 2),
        "all_scores": preds[0]
    }
```

Figure-1 : Code snippet of complete classification pipeline, including image preprocessing, normalization, and prediction using the EfficientNet model.

```python
def detect_changes_unet(image1, image2):
    """
    Change detection using U-Net model.
    Uses trained U-Net if available, otherwise falls back to image differencing.
    """
    global UNET_TRAINED

    diff_map, diff_percentage, diff_visualization = _compute_diff_change_map(image1, image2)

    if not UNET_TRAINED:
        return diff_map, diff_percentage, diff_visualization

    input_tensor = preprocess_for_unet(image1, image2)
    pred = unet_model.predict(input_tensor, verbose=0)[0]
    if pred.ndim == 3 and pred.shape[-1] == 1:
        pred = np.squeeze(pred, axis=-1)

    change_map = (pred > 0.5).astype(np.uint8)
    change_percentage = np.mean(change_map) * 100

    if not _is_valid_change_map(change_map):
        print(f"UNet prediction invalid (change_percentage={change_percentage:.2f}), using fallback differencing")
        return diff_map, diff_percentage, diff_visualization

    iou = _mask_iou(change_map, diff_map)
    if iou < 0.2 and abs(change_percentage - diff_percentage) > 15:
        print(f"UNet and diff disagree (iou={iou:.2f}, unet={change_percentage:.2f}%, diff={diff_percentage:.2f}%), using fallback diff
        return diff_map, diff_percentage, diff_visualization

    visualization = cv2.resize(image2, (256, 256))
    visualization[change_map == 1] = [255, 0, 0]
    return change_map, change_percentage, visualization
```

Figure-2: Code snippet demonstrating the change detection pipeline using a U-Net model..

```python
        message : "Failed to perform change detection"
    }), 500
def analyze_image_patches(image_rgb: np.ndarray, patch_size: int = 64, stride: int = 32):
    """Run sliding-window classification and compute per-class counts and coverage."""
    height, width, _ = image_rgb.shape
    class_counts = {name: 0 for name in CLASS_NAMES}
    total_patches = 0
    for y in range(0, max(height - patch_size + 1, 1), stride):
        for x in range(0, max(width - patch_size + 1, 1), stride):
            patch = image_rgb[y:y + patch_size, x:x + patch_size]
            if patch.shape[0] != patch_size or patch.shape[1] != patch_size:
                continue
            patch_float = patch.astype(np.float32) / 255.0
            pred = model.predict(np.expand_dims(patch_float, 0), verbose=0)
            class_idx = int(np.argmax(pred, axis=1)[0])
            class_name = CLASS_NAMES[class_idx]
            class_counts[class_name] += 1
            total_patches += 1

    coverage = {k: (v / total_patches * 100.0 if total_patches else 0.0) for k, v in class_counts.items()}
    return {
        'width': width,
        'height': height,
        'patch_size': patch_size,
        'stride': stride,
        'total_patches': total_patches,
        'class_counts': class_counts,
        'coverage_percent': coverage,
    }
```

Figure-3: Code snippet of sliding window approach to analyze large satellite images

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